

AEM Dialog Fields

Complete Reference Guide

Granite UI · Touch UI · JCR Properties · Sightly/HTL Usage

1. What is nt:unstructured?

Every node in AEM's JCR (Java Content Repository) must declare a **node type**. The node type acts like a schema — it controls what properties and child nodes are allowed. **nt:unstructured** is the most permissive node type: it has *no schema constraints*. You can add any property, any child node, in any combination.

Node type comparison

Node Type	Acts Like	Schema?	Common Use
nt:unstructured	Blank sticky note — anything goes	None	Dialogs, configs, component nodes
	A page with required jcr:content child	Strict	Content pages
	A file with required content node	Strict	Binary files in JCR
	DAM asset with metadata structure	Strict	Images, documents in DAM
	Folder that organises children	Moderate	Repository folders

Why dialogs use it

Dialog XML files are developer-authored configuration nodes — not end-user content. AEM doesn't need to enforce a schema here because developers control what goes in. Using **nt:unstructured** means AEM will never reject a property you add, making it the default choice for all Granite UI dialog nodes, design configs, and component definitions.

■ The {Boolean} syntax: In AEM XML, booleans must be written as {Boolean>true — writing plain true stores the String "true", not a JCR boolean. Same applies to {Long}42, {Double}3.14, etc.

2. Dialog XML Structure

Every Touch UI dialog is a **.content.xml** file placed at `_cq_dialog/.content.xml` inside your component folder. The outer wrapper tells AEM what kind of node this is; the inner fields define what authors can edit.

```
<jcr:root
```

```

jcr:primaryType="nt:unstructured"
sling:resourceType="cq/gui/components/authoring/dialog"
jcr:title="Card Settings">
<content sling:resourceType="granite/ui/.../container">
  <items jcr:primaryType="nt:unstructured">
    <!-- field nodes go here -->
  </items>
</content>
</jcr:root>

```

XML node / attribute	Role
jcr:primaryType="nt:unstructured"	Every dialog node uses this — no schema constraints
sling:resourceType="cq/gui/.../dialog"	Marks this as a Touch UI dialog (not Classic UI)
jcr:title	Title shown in the dialog window header bar
content > items	Granite UI container that holds all field widgets
name="./foo"	Stores value as property "foo" on the component's JCR node

3. Field Types Reference

TextField — single-line input

[granite/ui/components/coral/foundation/form/textfield](#)

Key attributes

```

name="./title"
fieldLabel="Title"
required="{Boolean}true"
maxlength="80"
fieldDescription="The main heading"

```

What each attribute does

name → JCR property (./ = relative to component node)
required → blocks Save until filled
maxlength → HTML maxlength on the input
fieldDescription → tooltip shown below the field
value → default value on new component instance

JCR node output

```

title = "My Card Title"
(stored as String property)

```

Sightly / HTL usage

```

${properties.title}

```

Validation fires on save

Single-line only

Textarea — multi-line input

[granite/ui/components/coral/foundation/form/textarea](#)

Key attributes

```
name="./description"
fieldLabel="Description"
rows="3"
```

What each attribute does

rows → sets visible height of the textarea
Newlines stored as \n in JCR
For bold/links/lists use richtext instead
No built-in maxlength UI (add JS if needed)

JCR node output

```
description = "Line one\nLine two"
(String – newlines preserved)
```

Sightly / HTL usage

```
${properties.description}
```

Newlines: use nl2br in HTL

PathField — page or asset browser

[granite/ui/components/coral/foundation/form/pathfield](#)

Key attributes

```
name="./ctaLink"
fieldLabel="Button Link"
rootPath="/content"
name="./imageRef"
rootPath="/content/dam"
```

What each attribute does

rootPath → where the browse tree opens
/content → for pages
/content/dam → for DAM assets
Author types OR clicks browse icon
Stores raw JCR path, NOT a URL
Use Externalizer/LinkChecker in Java
to convert to a proper URL

JCR node output

```
ctaLink = "/content/mysite/en/about"
imageRef = "/content/dam/hero.jpg"
(String – raw JCR paths)
```

Sightly / HTL usage

```
${properties.ctaLink}.html
```

No URL conversion auto

Supports autocomplete

Select — dropdown with fixed options

[granite/ui/components/coral/foundation/form/select](#)

Key attributes

```
name="./theme"
fieldLabel="Card Theme"
<items>
  <default jcr:title="Default"
    value=""
    selected="{Boolean}true"/>
  <dark jcr:title="Dark"
    value="theme-dark"/>
  <light jcr:title="Light"
    value="theme-light"/>
</items>
```

What each attribute does

jcr:title → label shown in dropdown
value → what gets stored in JCR
selected="{Boolean}true" → pre-selected default
For dynamic options use datasource child node
backed by a Sling model or servlet

JCR node output

```
theme = "theme-dark"
(String – the value attribute)
```

Sightly / HTL usage

```
<div class="${properties.theme}">
```

Static or dynamic options

Empty value = no property saved

Checkbox — boolean toggle

[granite/ui/components/coral/foundation/form/checkbox](#)

Key attributes

```
name="./openInNewTab"
fieldLabel="Open link in new tab"
value="true"
uncheckedValue="false"
```

What each attribute does

value → stored when checked
uncheckedValue → stored when unchecked
CRITICAL: without uncheckedValue, unchecking
DELETES the property from JCR entirely.
It does NOT store false — the property
simply disappears.
Always add uncheckedValue="false"!

JCR node output

```
Checked:  openInNewTab = "true"
Unchecked: openInNewTab = "false"
```

Sightly / HTL usage

```
${properties.openInNewTab == 'true'}
```

Without uncheckedValue → property deleted on uncheck

NumberField — numeric input

[granite/ui/components/coral/foundation/form/numberfield](#)

Key attributes

```
name="./priority"
fieldLabel="Priority"
min="1"
max="10"
step="1"
```

What each attribute does

min/max → validation at browser + Granite level
step → increment/decrement step size
Stored as String in JCR by default!
To store as Long integer, add a hidden field:
name="./priority@TypeHint"
value="Long"
Or cast in your Sling Model with @ValueMapValue

JCR node output

```
priority = "5"
(String unless @TypeHint used)
```

Sightly / HTL usage

```
${properties.priority}
```

Stored as String — parse in Java

Use @TypeHint for numeric JCR type

DatePicker — date / time picker

[granite/ui/components/coral/foundation/form/datePicker](#)

Key attributes

```
name="./expiryDate"
fieldLabel="Expiry Date"
type="datetime"
```

What each attribute does

type="date" → date only (YYYY-MM-DD)
type="datetime" → date + time picker
type="time" → time only
displayedFormat → what the author sees
e.g. DD/MM/YYYY HH:mm
valueFormat → what gets stored in JCR
Stored as ISO 8601 string with timezone

JCR node output

```
expiryDate = "2025-12-31T23:59:00.000+05:30"
(ISO 8601 String with timezone offset)
```

Sightly / HTL usage

```
${properties.expiryDate}
```

Timezone-aware ISO string

Parse with Java Instant or Calendar

4. Resulting JCR Node

After an author fills in all fields and clicks Done, AEM's Sling POST Servlet writes each name="./foo" value as a property on the component's JCR node. This is what the node looks like in CRXDE:

```
/content/mysite/en/home/jcr:content/par/card
- jcr:primaryType = "nt:unstructured"
- sling:resourceType = "mysite/components/card"
- title = "My Card Title"
- description = "Some descriptive text"
- ctaLink = "/content/mysite/en/about"
- imageRef = "/content/dam/mysite/hero.jpg"
- theme = "theme-dark"
```

```
- openInNewTab      = "true"
- priority          = "3"
- expiryDate        = "2025-12-31T23:59:00.000+05:30"
```

How the flow works

1

Author opens component

AEM reads the dialog XML and renders the Granite UI fields in the edit dialog.

2

Author fills fields & clicks Done

Browser POSTs all field values to the Sling POST Servlet at the component's JCR path.

3

Sling POST Servlet writes JCR

Each name="/foo" becomes a String property on the component node (unless @TypeHint overrides).

4

Page is rendered

Your HTL template reads properties.foo and outputs the values into HTML.

5. Common Gotchas



Checkbox without uncheckedValue

If uncheckedValue is missing and the author unchecks the box, the property is deleted from JCR — not set to false. Always add uncheckedValue="false".



PathField stores JCR path, not URL

PathField gives you /content/mysite/en/page — not a URL. Append .html in HTL or use Externalizer in Java to get a clickable link.



Numbers stored as Strings

All dialog values are posted as Strings. Use a hidden @TypeHint field or cast in your Sling Model to get Long, Double, etc.



{Boolean} vs "true"

In dialog XML, write {Boolean>true for actual JCR booleans. Writing true without the type hint stores the String "true".



Tabs in dialogs

Wrap field groups in a container with
sling:resourceType="granite/ui/components/coral/foundation/container/tabs" to add tabs.



Conditional show/hide

Use granite:hide expression or showHideTarget/showHideCondition attributes to show or hide fields based on other field values.

**Dynamic select options**

Instead of static , use a datasource child node backed by a Sling model to populate dropdown options from JCR or an API.

**Multifield (repeating rows)**

Use `granite/ui/components/coral/foundation/form/multifield` to let authors add multiple instances of a field group.